A Dual-Stream Transformer With Diff-Attention for Multispectral and Panchromatic Classification

Lin Xu, Hao Zhu, *Member, IEEE*, Licheng Jiao, *Fellow, IEEE*, Wenhao Zhao, Xiaotong Li, Biao Hou, *Senior Member, IEEE*, Zhongle Ren, *Member, IEEE*, and Wenping Ma, *Senior Member, IEEE*

***Abstract*— To minimize the feature redundancy of multispectral (MS) and panchromatic (PAN) images and maximize the complementary advantages of PAN and MS, a dual-stream transformer with diff-attention (DSTD)-Net is proposed for PAN and MS classification in this article. First, in terms of feature extraction, we use self-attention and coattention (SCA) block to extract both specific advantageous features and common essential features. Based on that, a self-attention module strengthened by diff-attention (SSDA) that pays attention to the difference between two specific advantageous features is designed to reduce the essential redundancy in specific features. It can take advantage of the difference between two specific features and reduce the essential redundancy of the specific advantageous features, making them purer and better for classification. Finally, since the specific features and common features of MS and PAN images make different contributions to classification, a multistage gated fusion strategy is used. The multistage gate fusion (MGF) strategy mainly uses gated multisource units (GMUs) to adapt the weight of different features and fuse them. So, our MGF strategy can strengthen the specific advantageous features beneficial for classification. Above all, the several experiment results verify our proposed networks' effectiveness and robustness. Our code is available at: https://github.com/blackkiring/DSTD.**

***Index Terms*— Attention mechanism, classification, deep learning (DL), feature fusion, remote sensing, transformer.**

I. Introduction

With the fast development of geospatial information technology, remote sensing satellites can more easily acquire images from different sources, such as multispectral (MS) and panchromatic (PAN) images. Nowadays, only using the data in a single modality is unable to achieve the desired performance. Since PAN images have high spatial resolution while MS images have high spectral resolution, we must find a suitable way to integrate the advantages of MS images and PAN images. In recent years, how to improve the classification performance of remote sensing images through multisource fusion has become a hot research issue.

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The authors are with the Key Laboratory of Intelligent Perception and Image Understanding of Ministry of Education, International Research Center for Intelligent Perception and Computation, Joint International Research Laboratory of Intelligent Perception and Computation, School of Artificial Intelligence, Xidian University, Xi'an 710071, China (e-mail: haozhu@xidian.edu.cn).

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In the remote sensing field, the major multisource classification algorithms can be divided into two categories: one is the traditional method, and the other is the algorithms based on deep learning (DL) [1], [2], [3], [4].

One of the traditional methods is to pan-sharp the MS images first and then classify them. These algorithms usually require suitable pan-sharping methods [5], [6], [7]. Meng et al. [6] review the traditional and latest pan-sharpening methods these years. These traditional pan-sharpening methods can be divided into three types [6]: the component substitution (CS)-based algorithms [8], multiresolution analysis (MRA)-based methods [9] and variational optimization (VO)-based methods [10]. The CS-based method is also called projection–substitution method, which can be separated into the principle component analysis (PCA) methods [11], [12], Gram-Schmidt (GS) methods [13] and the intensity-hue-saturation (IHS) methods [14]. In terms of MRA-based methods [15], there are two kinds of algorithms, including the high-pass filter (HPF) methods [16], [17] and the decimated wavelet transform (DWT) methods [18], [19]. The VO-based methods have two steps: constructing the energy function and optimizing the solution. In terms of the construction of the energy function, the methods based on the observation model [20] and the sparse representation are the most popular.

The other traditional method is the way of feature-fusion-then-classify, which extracts features separately from PAN and MS images. Jia et al. [21] raise a spectral-spatial Gabor surface feature fusion algorithm for hyperspectral imagery classification. Hedhli et al. [22] classify the multisensor and multiresolution remote sensing images by hierarchical Markov random fields.

In recent years, with the rapid development of the speed at which computers process data, the DL methods [23], [24], [25], [26] in the remote sensing field have become very popular. Zhang et al. [27] propose a single-source domain expansion network (SDEnet), which employs the supervised contrastive learning for classwise domain invariant representation and designs adversarial training to optimize the generator to separate intraclass samples of source domain (SD) and extended domain (ED). Zhang et al. [28] propose a structural optimization transmission network (SOT-Net), which takes advantage of reflectance-specific information of hyperspectral image (HSI) and edge representations of multisource data. Liu et al. [29] propose a scaled dot-product central attention (SDPCA) for the spectral and spatial information from the central pixel and similar pixels.

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Nowadays, along with Transformer's remarkable success in natural language processing, Dosovitskiy et al. [30] experimented with applying Transformer [31], [32] to vision. Transformer has been used in Remote Sensing [33], [34]. Su et al. [34] raise a Transformer-based regression network for pan-sharpening. Bandara and Patel [35] create a Hyper-Transformer, which uses a convolutional network to extract features from PAN and MS images and then uses a transformer model to construct a new image with both spatial and spectral features. Moreover, many networks with attention mechanisms [36], [37], [38] for remote sensing images are proposed. Zhou et al. [36] propose a novel multiple fusion model with both attention multihop graph and multiscale convolutional for HSI classification. Luo et al. [37] propose a bidirectional diff-changed feature representation (BDFR) module for HSI change detection. Guo et al. [38] combine dual-view spectral and global spatial features for HSI classification.

However, we still have some problems remaining to be solved. Here are the major challenges.

1) In the feature extraction stage, since MS and PAN images are data from different sources with their own advantages, we need a suitable feature extractor to extract common essential features as the bridge for feature fusion and specific advantageous features for classification. Furthermore, many researchers tend to pay more attention to reducing modal differences instead of removing redundant features. Both reducing modal differences and removing redundant features are important for final classification. A large amount of redundant information will overwrite the specific advantageous features if we don't reduce the redundancy in specific features. So, we should find a suitable approach to removing redundancy between specific advantageous features.
2) In the stage of feature fusion, the specific advantageous features and common essential features contribute differently to the final classification. Since the specific advantageous features are more important for classification and the common essential features are necessary for feature fusion, finding a suitable weight for specific advantageous features to fuse is significant.

Based on the above analysis of the above challenges, we propose a new structure of transformer for the classification based on the fusion of MS and PAN images, which is used to extract the common essential features of MS and PAN and their respective specific features. We define the surface details from different modalities that are beneficial for classification as the specific advantageous features. What is more, we define the high-level semantic information that represents what the object is as the common essential features. Our major contributions are the following.

1) In the feature extraction stage, we propose a new feature-extracting block with the attention mechanism. First, we use a self-attention and coattention (SCA) block guided by spatial and spectral attention maps to extract specific advantageous features and common essential features. Based on that, we design a self-attention module strengthened by diff-attention (SSDA) block. The SSDA block can pay attention to the difference between two specific advantageous features. Then, we take advantage of the difference between two specific features and reduce the essential redundancy of the specific advantageous features. Therefore, our specific advantageous features become purer, which is beneficial for classification.
2) In the stage of feature fusion, we introduce gated multisource units (GMUs) [39] to our model. Considering that the specific advantageous features and common essential features of MS and PAN have different contributions to the classification task, we propose our multistage gate fusion (MGF) strategy. Our MGF strategy can adaptively adjust the weight of different features according to their different contributions to classification. So, our MGF strategy can strengthen the specific advantageous features that are beneficial for classification and weaken the common features in the stage of feature fusion.

II. Related Work

In this section, we introduce some related works of our models in detail.

A. Visual Transformer

Visual transformer (ViT) [30] is inspired by the field of natural language processing. Compared with the convolution network, the transformer is better at fetching long-distance dependencies and extracting global features. In ViT, Patch Embedding is used to convert the raw 2-D image into a series of 1-D patch embeddings. If defining the input images and the patch-embedding operation as $x \in \mathbb{R}^{b\times c\times w\times h}$ and f_{patch}, we can get the equation as follows:

$$x_{\text{in}} = f_{\text{patch}}(x) + x_{\text{pos}}$$

where $x_{\text{in}} \in \mathbb{R}^{b\times N\times (P^2C)}$ and $x_{\text{pos}} \in \mathbb{R}^{b\times N\times (P^2C)}$. The N is the number of image blocks being cut, and P is the size of each small patch. The operation of f_{patch} cuts the images into many small patches by a linear layer and then stretches each small patch into a 1-D vector. The x_{pos} is a set of learnable location codes with the same size as $f_{\text{patch}}(x)$. Self-attention is one of the most important components of the Transformer. In the self-attention module, W_q, W_k, and W_v are a group of learnable matrices, and then we define $Q := xW_q$, $K := xW_k$, $V := xW_v$

$$\text{SA}(x) = \frac{\text{softmax}\left(QK^T\right)V}{\sqrt{d_k}} \tag{1}$$

$$\text{head}_i(x) = \frac{\text{softmax}\left(Q_iK_i^T\right)V_i}{\sqrt{d_k}} \tag{2}$$

$$\text{MHSA}(x) = \text{Concat}(\text{head}_1, \text{head}_2, \ldots,)W_O \tag{3}$$

where d_k is the dimension of vectors, which is a scale factor to avoid gradient descent caused by the final value being too large or too small. In multihead self-attention, different attention heads project Q, K, and V into different subspaces, and the regions of subattention are different in different subspaces, making the model more generalizable. However, because of

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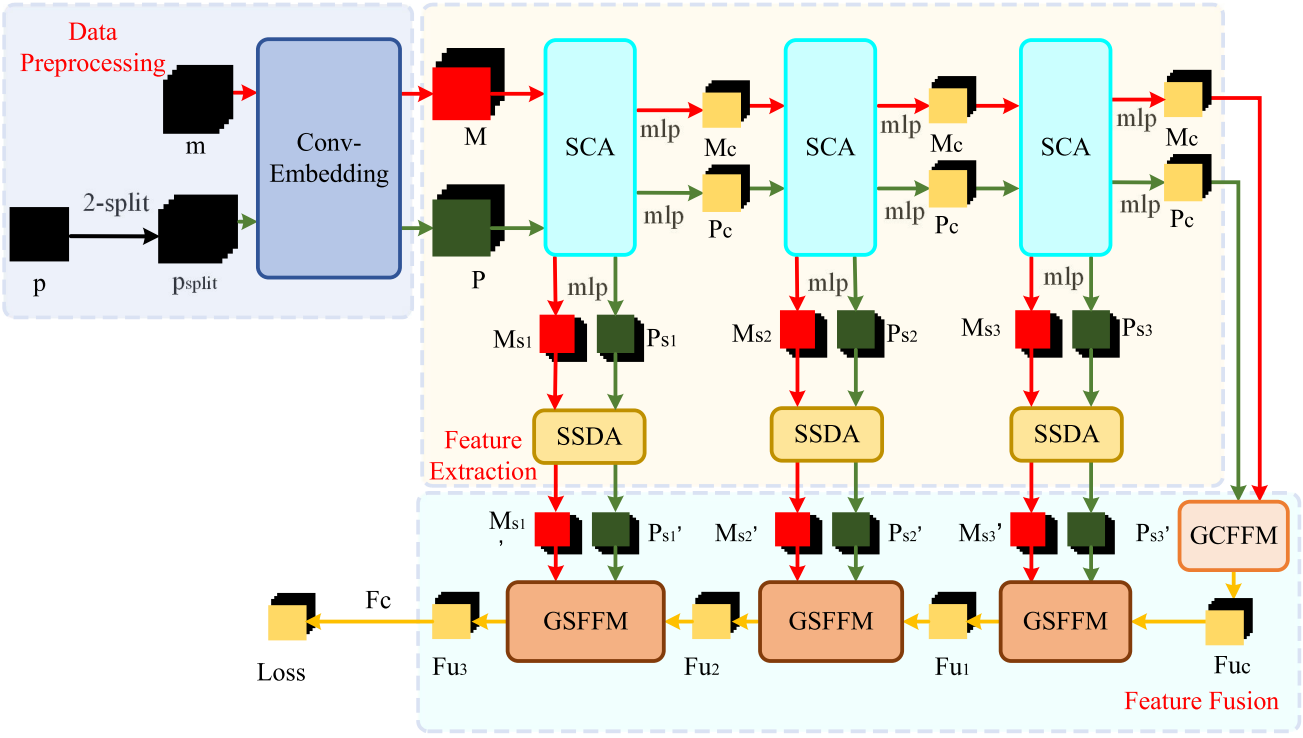


Fig. 1. Whole structure of DSTD. The whole process consists of three parts: 1) the data processing stage; 2) the feature extraction stage with the SCA block and the SSDA block; and 3) the feature fusion stage with MGF strategy. The MGF strategy includes two parts: the GCFFM and the GSFFM.

the lack of inductive bias, ViT isn't able to perform very well in small datasets.

Wu et al. [40] present a new architecture named Convolutional vision Transformer (CvT). Different from the ViT, CvT uses a convolutional embedding layer to replace the patch embedding layer. Compared with the patch embedding layer, the convolutional token embedding layer models local context information. With the help of the convolutional token embedding layer, it is possible for us to gradually reduce the length of the token sequence while increasing the feature dimension of each token. Introducing this convolutional layer allows the transformer to extract features from different stages, like convolutional neural networks, and incorporate local features. This work also replaces the linear layer with the convolutional projection for the form of Q, K, and V

$$x^{q/k/v} = \text{Flatten}(\text{Conv2d}(\text{Reshape2d}(x), s)). \tag{4}$$

III. Our Method

This article proposes a novel structure of transformer for the classification based on the fusion of MS and PAN images, which is used to extract the common essential features and respective specific advantageous features of the MS image and the PAN image. The whole structure of our dual-stream transformer with diff-attention (DSTD) is shown in Fig. 1. Our transformer mainly includes three parts: the data processing stage, the feature extraction stage with the SCA block and the SSDA block, and the feature fusion stage with MGF strategy.

A. Data Preprocessing

Since the MS image and PAN image have different resolutions, we need to preprocess them for convenience. In Fig. 1, $p \in \mathbb{R}^{4w\times 4h\times 1}$, and $m \in \mathbb{R}^{w\times h\times 4}$ represent the PAN image and MS image. We use the approach of two-split [26] to convert the PAN image from single-channel image to four-channel image $p_{\text{split}} \in \mathbb{R}^{2w\times 2h\times 4}$. The two-split approach divides the PAN image into 2×2 patches and extracts the pixels at the same position in each patch as a channel of the new image. The details are shown as follows:

$$p_{\text{split}-i}(x, y) = p(2x - 1 + (i-1)\%q, 2y - 1 + \lfloor (i-1)/q \rfloor)$$
$$i = 1, 2, 3, 4, \quad x = 1, \ldots, 2w, \quad y = 1, \ldots, 2h. \tag{5}$$

Like most posted ViTs [30], [31], [32], [34], [41], our transformer also has the patch-embedding model. Our patch-embedding model uses several 3×3 overlapping convolution layers, which are used to strengthen the connection between patches and serve as an implicit spatial location encoding. The overlapping convolution layers also include an operation of downsampling, which can reduce the time of this module. We define the operation of patch-embedding as f_{pe}, so the outputs of f_{pe} are defined as P and M

$$P, M = f_{\text{pe}}(p_{\text{split}} \downarrow, m) \tag{6}$$

where $P \in \mathbb{R}^{C\times(w/2)\times(h/2)}$ and $M \in \mathbb{R}^{C\times(w/2)\times(h/2)}$. For simplicity, we discard the batch dimension from our notations. The C represents the number of channels of the features. The w and h represent the weight and height of our feature maps.

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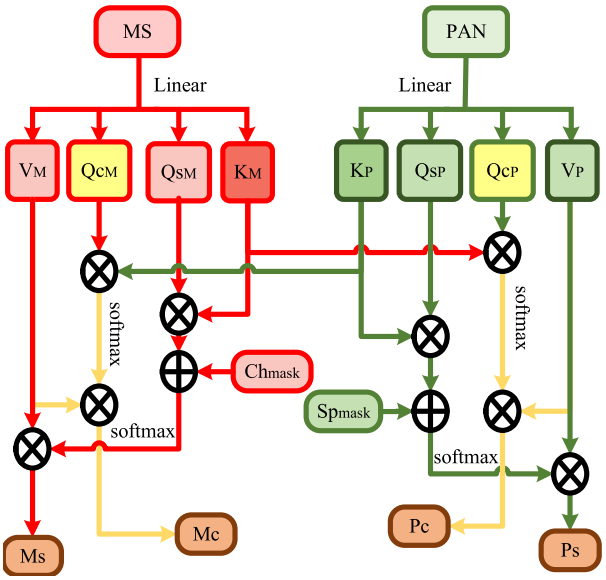


Fig. 2. SCA block. $\mathrm{Ch}_{\mathrm{mask}}$ and $\mathrm{Sp}_{\mathrm{mask}}$ are the two masks that help the block focus on the specific advantageous features. Mc and Pc are the common essential features, and Ms and Ps are the specific advantageous features.

B. Self-Attention and Coattention Block

Recently, traditional transformer algorithms have been used to classify single-source data. However, with the development of technology, we know that the application of data from more than one source is beneficial for classification. So it is of great potential for us to improve the transformer and adapt it to the remote sensing classification tasks in a dual-source context. Some researchers [35], [42] use the coattention module in the transformer to fuse specific advantageous features. But, in their networks, the specific advantageous features lack interactions, which will cause the redundancy of the essential feature. Therefore, we design the SCA block as follows.

In our SCA block, the self-attention mechanism is used to extract PAN and MS images' respective specific advantageous features while the coattention mechanism extracts their common essential features by computing the similarity of their features. The structure of our attention module is shown in Fig. 2.

First, we use a linear layer to separately map MS feature and PAN feature to four features—self-Query (Qs), co-Query (Qc), Key (K), and Value (V) features instead of the traditional three maps—K, Q, and V. We don't use the traditional operation because the feature Q will be used in both SCA, which we think causes the fact that the common essential features and specific advantageous features cannot be split very well. So, we define the above operation as $f_{\mathrm{linear}}(\cdot)$, and then we can define Qs, Qc, K and V as follows:

$$K_M, \mathrm{Qs}_M, \mathrm{Qc}_M, \quad V_M = f_{\text{linear-}m}(M) \tag{7}$$

$$K_P, \mathrm{Qs}_P, \mathrm{Qc}_P, \quad V_P = f_{\text{linear-}p}(P) \tag{8}$$

where $K, \mathrm{Qs}, \mathrm{Qc}$, and V are all the transformations of MS/PAN feature. The four features from the same source essentially have similar properties. After we get the elements the attention mechanism needs, we need to compute the SCA masks, which are used to extract the specific advantageous features and common essential features separately.

The formations of attention maps are shown as follows:

$$\mathrm{SA}_M = \mathrm{softmax}\left(\frac{\mathrm{Qs}_M \times K_M^T}{\sqrt{d_k}} + \alpha \times \mathrm{Ch}_{\mathrm{mask}}\right) \tag{9}$$

$$\mathrm{SA}_P = \mathrm{softmax}\left(\frac{\mathrm{Qs}_P \times K_P^T}{\sqrt{d_k}} + \beta \times \mathrm{Sp}_{\mathrm{mask}}\right) \tag{10}$$

$$\mathrm{CA}_M = \mathrm{softmax}\left(\frac{\mathrm{Qc}_P \times K_M^T}{\sqrt{d_k}}\right) \tag{11}$$

$$\mathrm{CA}_P = \mathrm{softmax}\left(\frac{\mathrm{Qc}_M \times K_P^T}{\sqrt{d_k}}\right) \tag{12}$$

where SA_M, SA_P, CA_M, and CA_P are all the attention masks. Each element in the attention maps means the strength of connections between two features. The more similar the two features, the greater their corresponding values in the attention masks. The SA_M and SA_P majorly focus on MS and PAN images' specific advantageous features, while CA_M and CA_P pay more attention to their common essential features. MS images have multiple spectral bands containing rich spectral information, while PAN images have high spatial resolution and rich spatial information. Thus the specific advantageous features of MS images should pay more attention to spectral information, and PAN images should focus more on spatial information. So we set the $\mathrm{Ch}_{\mathrm{mask}}$ and $\mathrm{Sp}_{\mathrm{mask}}$ as a group of masks to guide our features to their specific advantageous features. Furthermore, we think that the specific advantageous features not only include spectral information or spatial information but also have some other high-level advantageous features. The suitable weight parameters are required for $\mathrm{Ch}_{\mathrm{mask}}$ and $\mathrm{Sp}_{\mathrm{mask}}$ to avoid the two masks overriding other useful information for classification. So we use α and β as the weight parameters of $\mathrm{Ch}_{\mathrm{mask}}$ and $\mathrm{Sp}_{\mathrm{mask}}$. Here are the formations of $\mathrm{Ch}_{\mathrm{mask}}$ and $\mathrm{Sp}_{\mathrm{mask}}$

$$\mathrm{Ch}_{\mathrm{mask}} = \frac{\text{SE-Net}(\mathrm{Qs}_M) \times K_M^T}{\sqrt{d_k}} \tag{13}$$

$$\mathrm{Sp}_{\mathrm{mask}} = \frac{\mathrm{SpA}(\mathrm{Qs}_P) \times K_P^T}{\sqrt{d_k}} \tag{14}$$

where $\mathrm{Ch}_{\mathrm{mask}}$ and $\mathrm{Sp}_{\mathrm{mask}}$ are two guided masks that pay attention to spectral and spatial features. Since the SE-Net [43] and spatial attention (SpA) [44] are able to extract the spatial features and spectral features well, we apply the two methods to form our guiding masks. With the guidance of two masks, our self-attention is more explainable, and it provides a more appropriate direction for the self-attention to accelerate model convergence. Without the two masks, the self-attention may focus more on the global information and ignore some important local information, thus causing a worse performance. After fetching the attention maps, we can get their own common essential features and specific advantageous features by following equations:

$$M_s = \mathrm{SA}_M \times V_M \tag{15}$$

$$M_c = \mathrm{CA}_M \times V_M \tag{16}$$

$$P_s = \mathrm{SA}_P \times V_P \tag{17}$$

$$P_c = \mathrm{CA}_P \times V_P. \tag{18}$$

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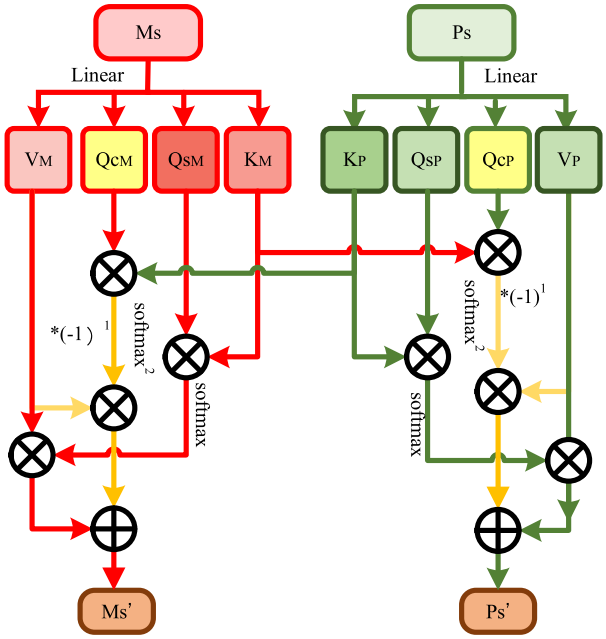


Fig. 3. SSDA. Ms′ and Ps′ are the purer specific advantageous features. The superscript in the figure indicates the order of operations.

Above all, our SCA block is able to extract both specific advantageous features and common essential features. Furthermore, the SCA block can focus on PAN and MS images' advantageous features with the help of SpA mask and spectral attention mask.

C. Self-Attention Module Strengthened by Diff-Attention

We think that, although self-attention can obtain PAN and MS images' respective specific advantageous features, self-attention only interacts with its own features and exists some essential feature redundancy. It is still required for us to use the difference between the two modalities to remove the intrinsic redundancy features. So, we design the strengthened self-attention module called SSDA as follows, aimed to find the difference between MS image and PAN image. Our SSDA block is shown in Fig. 3. Then we use followed equations to obtain diff-attention maps:

$$\mathrm{DA}_M = \mathrm{softmax}\left(-\frac{\mathrm{Qc}_P \times K_M^T}{\sqrt{d_k}}\right) \tag{19}$$

$$\mathrm{DA}_P = \mathrm{softmax}\left(-\frac{\mathrm{Qc}_M \times K_P^T}{\sqrt{d_k}}\right) \tag{20}$$

$$M_s' = \mathrm{SA}_M \times V_M + \mathrm{DA}_M \times V_M \tag{21}$$

$$P_s' = \mathrm{SA}_P \times V_P + \mathrm{DA}_P \times V_P \tag{22}$$

where we use the minus sign in (19) and (20) to convert the similarity matrix in coattention to the difference matrix. So that we are able to focus more on the elements that have smaller values, and they are also the elements that are more different.

Furthermore, we use both self-attention and diff-attention to compute specific features to minimize the similarity redundancy in the specific advantageous features of the two image sources. Compared with the traditional self-attention equations [see (15) and (17)], we add the diff-attention masks (DA_M and DA_P) to self-attention. So we can get the purer specific features M_s' and P_s' in (21) and (22). Our idea of diff-attention is inspired by coattention. In coattention, K and Q are from different sources, and they use the dot-product attention to calculate the similarity matrix [see (11) and (12)]. However, in our diff-attention, we calculate the difference matrix [see (19) and (20)]. Thus, in the diff-attention map, the less similar the two elements, the smaller the value corresponding to the attention matrix. All in all, the diff-attention is able to enhance features with weak correlations between two sources and reduce redundancy between two specific features.

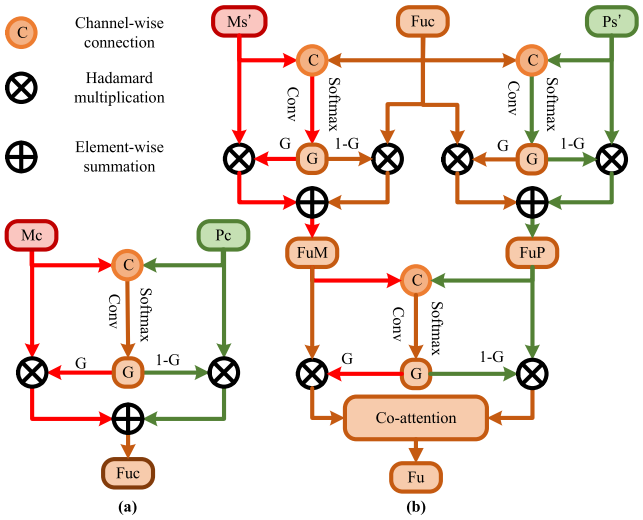


Fig. 4. Composition of MGF module. (a) GCFFM [39]. (b) GSFFM.

The traditional self-attention only depends on itself, however, our attention mechanism not only focuses on the information about the modality itself but also pays more attention to the different information between the two modalities (MS and PAN). Thus, the difference in information could make the specific advantageous features of the two sources purer and more discriminating.

D. Multistage Gate Fusion Strategy (MGF)

Since the specific advantageous features and common essential features of MS images and PAN images contribute differently to the final prediction result, we propose the MGF strategy to adjust the weights of different features adaptively. The GMU [39] learns to decide how modalities influence the unit's activation using multiplicative gates, which can find an intermediate representation based on a combination of data from different modalities. Therefore, we introduce the GMU, in our MGF strategy. First, we use the GMU to merge the common essential features of the two branches of our network. Fig. 4 shows the composition of our fusion model. Here are the equations of our model

$$G(x, y) = \mathrm{sigmoid}(\mathrm{conv}(\mathrm{cat}(x, y))) \tag{23}$$

$$F_{\mathrm{cu}}(x, y) = G \times x + (1 - G) \times y \tag{24}$$

$$F_{\mathrm{su}}(x, y) = \mathrm{CoA}(G \times x, (1 - G) \times y) \tag{25}$$

where x and y are the two features that need to be fused. G is the gate that adjusts the weights of the ensemble.

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Equations (23)–(25) are three functions we need to use in the following equations. Our MGF strategy comprises the gated common features fusion module (GCFFM) and the gated specific features fusion module (GSFFM). First of all, we use the GCFFM for the fusion of essential features (Mc and Pc) as the equation follows:

$$\mathrm{Fu}_c = F_{\mathrm{cu}}(\mathrm{Mc}, \mathrm{Pc}). \tag{26}$$

Since the specific features (Ms′ and Ps′) represent the dominant features of the respective modes and there are large modal differences between Ms′ and Ps′, it is unsuitable for us to fuse them together directly. So we set the output of the common features (Fu_c) as the bridge. Then the specific features (Ms and Ps) are fused separately with Fuc and get the two fused features FuM and FuP as the following equations show:

$$\mathrm{FuM} = F_{\mathrm{cu}}\left(M_s', \mathrm{Fu}_c\right) \tag{27}$$

$$\mathrm{FuP} = F_{\mathrm{cu}}\left(P_s', \mathrm{Fu}_c\right). \tag{28}$$

Finally, we use our gated fused module to fuse FuM and FuP

$$\mathrm{Fu} = F_{\mathrm{su}}(\mathrm{FuM}, \mathrm{FuP}) \tag{29}$$

where Fu is the final output of the GSFFM. The MGF strategy can adaptively select appropriate feature weights for fusion [see (23) and (24)], so that we can preserve specific features while using common features to reduce the semantic gap between features. What's more, we add a coattention mechanism to the gated fusion module in (25). The coattention is able to strengthen the correlation between FuM and FuP that remained to be fused, thus bridging the gap between two features. The stage of feature fusion uses the output of specific features of different transformer layers, thus we can ensure that the specific advantageous features of MS and PAN images are sufficiently extracted.

All in all, since the specific advantageous features have modal difference, we can use the common essential features as the bridge and add the coattention mechanism to reduce the modal difference. What's more, the idea of GMU [39] is able to adaptively adjust the weight of different features, so we can strengthen the specific advantageous features that are beneficial for classification, and avoid them overwritten by redundant common essential features.

E. Loss Function

Our loss function includes two parts: the feature separation loss and the cross-entropy loss function. The feature separation loss is used to separate common essential features and specific advantageous features. The equation of loss is shown as follows:

$$\begin{aligned} L1 = {} & \mathrm{softmax}(\mathrm{sim}(M_c, P_c)) \\ & + \mathrm{softmax}(-\mathrm{sim}(P_c, P_{s3})) + \mathrm{softmax}(-\mathrm{sim}(M_c, M_{s3})) \end{aligned} \tag{30}$$

$$L = L1 + \mathrm{Cross}(\mathrm{Fc}(\mathrm{Fu}_3), \mathrm{Label}). \tag{31}$$

IV. Experiment Results

A. Datasets Description

In this section, four datasets are used to verify the stability and robustness of our proposed model. All of the datasets include a pair of MS and PAN images, which are obtained for the GaoFen-2 satellite. For the GaoFen-2 satellite, the MS image has a spatial resolution of 3.2 m, and the PAN image has a spatial resolution of 0.8 m. Here are the details of these four datasets.

1) Showed in Fig. 5(a) is the Beijing area, whose MS image has 7300 × 6905 × 4 pixels, and PAN image owns 29200 × 27620 pixels. In this area, there are ten kinds of scenes: water, road, bare land, two types of vegetation that in different heights and five types of building areas.
2) Showed in Fig. 5(b) is the Hohhot area, whose MS image has 2001 × 2101 × 4 pixels, and PAN image owns 8004 × 8404 pixels. In this area, there are 11 kinds of scenes: water, road, bare land, vegetation, farmland, and six different types of building areas with different roof colors or in various areas.
3) Showed in Fig. 5(c) is the Xi'an area, whose MS image has 4548 × 4544 × 4 pixels, and PAN image owns 18192 × 18164 pixels. In this area, there are 12 kinds of scenes: water, bare land, low vegetation, farmland, tree, two types of roads and five types of building areas.
4) Showed in Fig. 5(d) is the Shanghai area, whose MS image has 7300 × 6905 × 4 pixels, and PAN image owns 29200 × 27620 pixels. In this area, there are ten kinds of scenes: water, road, bare land, two species of vegetation that come from different heights and five types of building areas that vary with each other.

B. Experimental Environment and Settings

Our proposed new transformer network is trained on a work-station with RTX3090 24-GB GPU and 128-GB random access memory (RAM) under Ubuntu 16.04 LTS. All the experiments of our network are based on the framework called PyTorch.

For evaluating the performance and rationality of our proposed network, we select three indicators: overall accuracy (OA), average accuracy (AA) and kappa statistics (KAPPA) to digitize the network classification results quantitatively. The patch size of both MS and PAN images as the input of our network is 32 × 32 × 4. In the network training stage, no more than 2% of labeled data are selected from each class as samples for training. If some classes have no more than 1500 samples, we will supplement the samples up to 1500 for balance. Moreover, samples whose intersection over union (IOU) ratio is greater than 1/3 with other training samples will be excluded from the test dataset. We set the initial learning rate as 1.2×10^{-4} and the batch size as 64. To reduce the influence of random parameters and guarantee the validity of the results, we repeated the experiment five times each time and used the mean as the final result. The parameters of our proposed DSTD-Net are shown in Table I.

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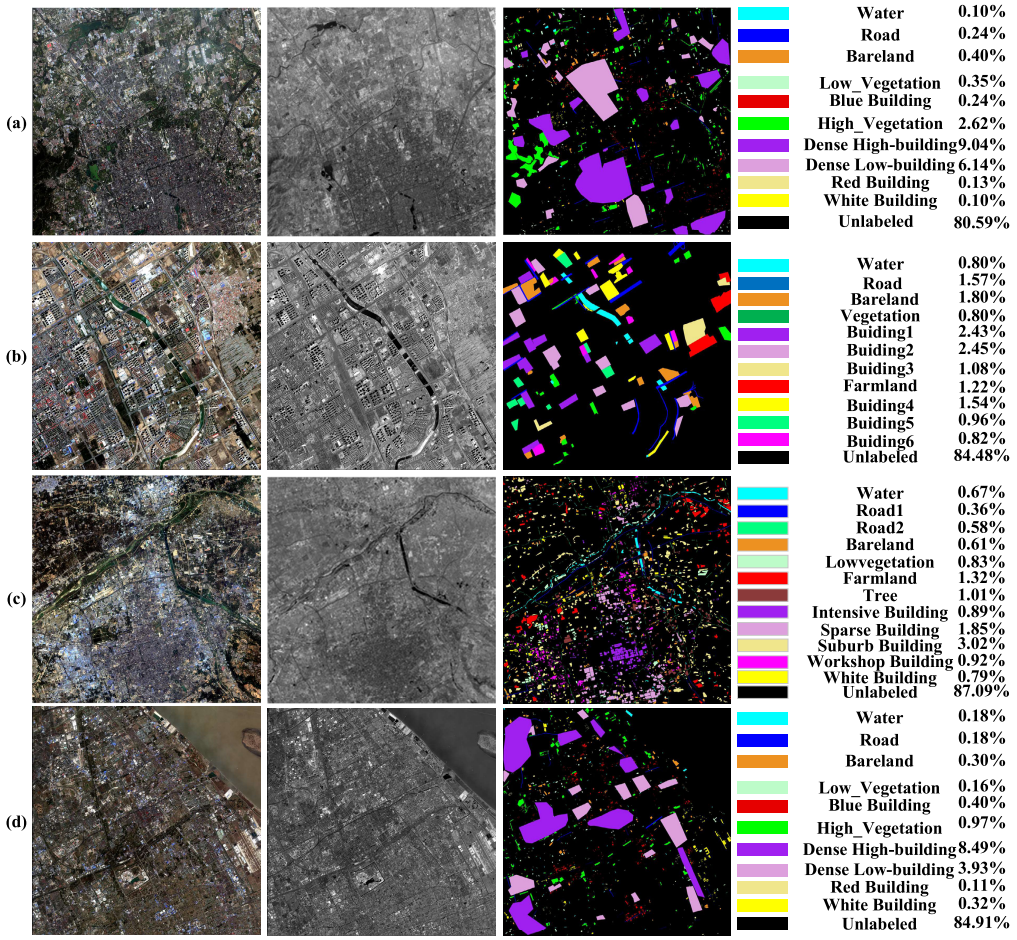


Fig. 5. Four rows in order from top to bottom are (a) Beijing area datasets, (b) Huhhot area datasets, (c) Xi'an area datasets, and (d) Shanghai area datasets. The four columns from left to right are the MS images, PAN images, ground-truth images, and corresponding labels. In addition, the MS images only display three channels, and the PAN images are resized to the same size of MS.

C. Hyperparameter Analysis and Selection

In this section, we analyze the hyperparameter detailedly and quantitatively using the control variable method. We keep the parameters the same except for the hyperparameters that selected.

1) *Experiment of the Patch Size:* As is shown in Table II, when the patch size is smaller, despite the fact that the needed training time is shorter, the performance of the network is worse. However, when the patch size is too large, the performance of our network is also not well. In fact, the situation is totally related to the resolution of the image and the size of the object. When the patch size is too small, the image cannot show the whole object. But when the patch size is too large, the image will include much invalid information that influences the classification performance. Considering the accuracy of our network, the patch size of 32 × 32 is the best and most effective.
2) *Experiment of the Transformer Block Number N:* In Table III, when the Transformer block number is small, there is no doubt that the performance of our network is worse. When the number of transformer layers is large, the performance in the major dataset is not very well; however, in some other datasets, the OA is improved. In our opinion, if the number of transformer blocks is too small, the network isn't complex enough to express the deep feature of the image. When the number of transformer blocks is too large, it takes too much time to train the network. Furthermore, it will cause the result of overfitting, which will make the performance of the test dataset worse. So we use three transformer blocks for training.

D. Module Comparison Experiments

In our module comparison experiment, we mainly compare the self-attention module with our SSDA module, and the gate fusion module with our proposed GSFFM on the randomly selected Beijing image.

1) *Self-Attention Module Versus SSDA:* Our SSDA block is better than the Self-attention Module (see Table IV), which is resulted from the fact that the Self-attention module only uses the features from one source and exists some common feature redundancy. The specific features extracted from the Self-attention module are not pure enough for better classification. But our proposed SSDA module not only considers the specific features from one modal but also considers the difference between the two specific features from different sources. So we can get purer specific features that are better for classification.

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TABLE I
HYPERPARAMETERS OF THE DTST-NET

Stage	Type	Input	PAN_Branch	MS_Branch	Output
Data Preprocess	prejust	MS(W×H,4) PAN(4W×4H,1)	$2split$, $4W \times 4H \times 1 \rightarrow 2W \times 2H \times 2$	None	$2W \times 2H$, 1 $W \times H$, 4
	Patch cutting	$2W \times 2H$, 2 $W \times H$, 4	None	None	64×64, 1 32×32, 4
Patch Embedding	None	64×64,1 32×32, 4	$\begin{bmatrix} \begin{bmatrix} conv2d: 4 \times 4 \\ Batchnorm \\ Relu \end{bmatrix} \times 2 \\ flatten + transpose \end{bmatrix}$	$\begin{bmatrix} \begin{bmatrix} conv2d: 3 \times 3 \\ Batchnorm \\ Relu \end{bmatrix} \times 2 \\ maxpool: 3 \times 3 \\ flatten + transpose \end{bmatrix}$	P (Ps_0): 256, 64 M (Ms_0): 256, 64
The ith Transformer block	Encode	Ms_{i-1}: 256, 64 Ps_{i-1}: 256, 64	$\begin{bmatrix} SCAM: \text{num-hand} = 2 \\ fc1: 64 \times 64 \\ Layernorm \\ fc2: 64 \times 64 \end{bmatrix}$		Ms_i: 256, 64 Ps_i: 256, 64 Mc_i: 256, 64 Pc_i: 256, 64
(i=1, 2, 3)	Decode	Ms1:256,64 Ps1:256,64	$\begin{bmatrix} Layernorm \\ SSDA: \text{num-head} = 2 \end{bmatrix}$	$\begin{bmatrix} Layernorm \\ SSDA: \text{num-head} = 2 \end{bmatrix}$	Ms1:256,64 Ps1:256,64
Fusion	GCFFM	Mc_3: 256, 64 Pc_3: 256, 64	$\begin{bmatrix} concat \\ fc: 128 \times 64 \\ gatefusion \end{bmatrix}$		$fu: 256, 64$
	GSFFM i (i=1, 2, 3)	Ms_{4-i}: 256, 64 Ps_{4-i}: 256, 64 fu: 256, 64	$\begin{bmatrix} concat: (Ms_{4-i}, fu), (Ps_{4-i}, fu) \\ fc: 128 \times 64 \\ gatefusion: fu1, fu2 \\ concat: fu1, fu2 \\ fc2: 128 \times 64 \\ gatefusion + \text{co-attention} \end{bmatrix}$		$fu: 256, 64$
Predict	None	256,64	$\begin{bmatrix} \text{attention-pool} \\ fc: 64 \times classes \end{bmatrix}$		classes

TABLE II
EXPERIMENT RESULT OF HYPERPARAMETER PATCH SIZE (%)

Dataset	Hohhot Area			Xi'an Area			Beijing Area			Shanghai Area		
Patch size	24	32	40	24	32	40	24	32	40	24	32	40
OA	99.32	99.42	99.19	98.95	99.64	98.92	88.3	94.74	95.52	88.2	95.35	95.33
AA	99.27	99.43	99.12	98.65	99.49	98.54	88.2	94.62	93.32	87.1	94.65	92.21
KAPPA	99.23	99.34	99.09	98.81	99.59	98.78	83.98	92.74	83.21	91.75	92.69	93.71

TABLE III
EXPERIMENT RESULT OF HYPERPARAMETER TRANSFORMER BLOCK NUMBER N (%)

Dataset	Hohhot Area			Xi'an Area			Beijing Area			Shanghai Area		
Transformer Block	2	3	4	2	3	4	2	3	4	2	3	4
OA	99.29	99.42	99.33	98.97	99.64	99.13	92.53	94.74	93.18	93.34	95.35	94.51
AA	99.30	99.43	99.24	98.78	99.49	98.77	92.17	94.62	90.23	92.17	94.65	91.28
KAPPA	99.21	99.34	99.25	98.85	99.59	99.01	89.58	92.74	90.51	90.21	92.69	91.93

2) *Gate Fusion Module Versus GSFFM:* The Gate Fusion Module is able to adaptively adjust the weights of specific features and common features since they have different contributions to classification. However, the modal gap of the specific features will disturb the performance of classification. Compared with the Gate Fusion module, our proposed GSFFM in Table V uses the coattention mechanism to reduce the difference between the two modalities, so we can better fuse the common features and specific features and have a better performance for classification.

E. Proof of Reasonableness

To prove the rationality of our network, we visualize our feature maps to see whether we can fetch the specific

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TABLE IV
SELF-ATTENTION VERSUS SSDA ON THE BEIJING IMAGE

Modules	self-attention	SSDA
c1:water	95.83	99.16
c2:road	91.48	94.44
c3:bare land	94.58	95.43
c4:low vegetation	93.37	94.48
c5:blue building	86.78	93.95
c6:high vegetation	97.18	96.68
c7:Dense high building	92.81	92.93
c8:Dense low building	96.09	97.01
c9:red building	88.22	87.76
c10:white building	93.26	93.34
OA(%)	94.02	94.74
AA(%)	92.96	94.62
Kappa(%)	91.73	92.74

TABLE V
GATE FUSION MODULE VERSUS GSFFM ON THE BEIJING IMAGE

Modules	Gate Fusion Module	SGFM
c1:water	91.58	99.16
c2:road	93.58	94.44
c3:bare land	96.74	95.43
c4:low vegetation	92.09	94.48
c5:blue building	91.53	93.95
c6:high vegetation	91.73	96.68
c7:Dense high building	92.97	92.93
c8:Dense low building	96.72	97.01
c9:red building	88.49	87.76
c10:white building	95.83	94.34
OA(%)	93.84	94.74
AA(%)	93.13	94.62
Kappa(%)	91.47	92.74

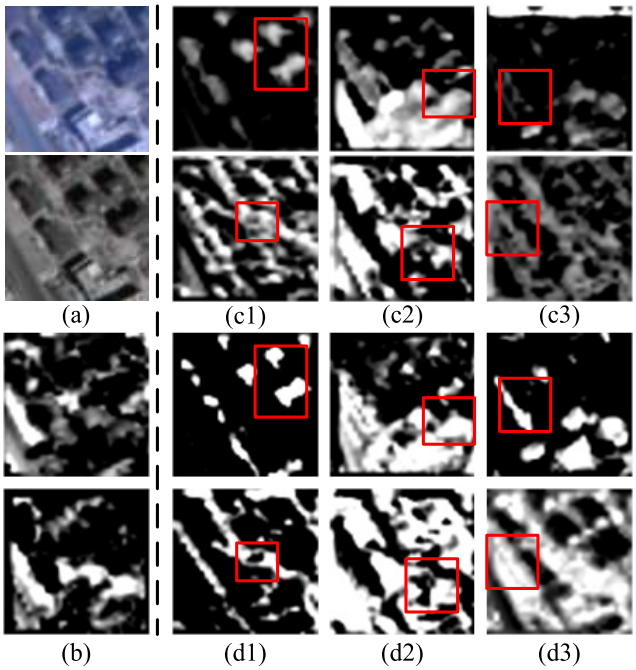


Fig. 6. Feature maps. (a) Input images (MS and PAN), where MS image displays three channels and PAN image is resized to the same size of MS. (b) Common features of MS and PAN. (c1)–(c3) Specific features without diff-attention. (d1)–(d3) Specific features with diff-attention.

TABLE VI
ABLATION EXPERIMENT RESULTS OF EACH COMPONENT ON THE BEIJING AREA DATASET (%)

Baseline	SCA*	SCA	SSDA	MGF	OA	AA	KAPPA
✓					84.5	80	77.91
✓	✓				85.97	87.94	80.9
✓		✓			86.67	83.79	81.43
✓			✓		85.32	88.86	80.16
✓	✓		✓		89.16	86.01	85.01
✓	✓			✓	88.28	87.40	83.76
✓		✓	✓		91.09	88.85	86.93
✓		✓		✓	92.33	90.12	88.56
✓		✓	✓	✓	**94.74**	**94.62**	**92.74**

advantageous features and common essential features in Fig. 6.

In Fig. 6(b), the common features are very similar and mainly include the same background information since common features are essential information. Fig. 6(c1)–(c3) are the information on the three channels that are specific features. The first line images of Fig. 6(c1)–(c3) are specific features of MS, so we can see that the three images strengthen different areas, such as roads and different buildings. However, in the second line, the specific features of PAN mainly focus on the road edge details. Furthermore, from the images of Fig. 6(d1)–(d3), we find that the specific features with diff-attention are clearer and purer than before. So we can say that our network is reasonable.

F. Ablation Study

In Table VI, "SCA*" is the SCA block that only uses Qsm and Qsp, while "SCA" is the complete SCA-block with Qcm and Qcp. As is shown in Table VI, the outcomes of the model with SCA block are all higher than those of the model with SCA* block. So, we think Qcm and Qcp are necessary for our model. The AA of the model with SSDA block improves a lot (88.86/83.79) from the model with SCA block. The SSDA block raises the stability of classification. What's more, the gap between the features is too large with the MGF strategy for feature fusion, resulting in little improvement. Above all, the results of DSTD-Net composed of SCA-block ∪ SSDA-block ∪ MGF are the highest.

G. Experimental Result With Beijing Area Dataset

This section compares our proposed network DSTD-Net with five other optimal algorithms. These six algorithms include spatial-channel progressive fusion (SCPF)-ResNet [49], spatial-channel collaborative attention (SCCA)-Net [45], an adaptive hybrid fusion network (AHF)-Net [2], multiscale progressive collaborative attention (MPCA)-Net [46], transfer representation learning fusion network (TRLF)-Net [47] and a global–local transformer network (GLT-Net) [48]. All parameters are set to the best parameters in their original paper to keep the experiments fair.

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TABLE VII
QUANTITATIVE RESULT ON THE BEIJING IMAGE

	SCPF-ResNet	SCCA-Net	AHF-Net Net	MPCA-Net	TRLF-Net	GLT-Net	DSTD-Net
c1	84.07	85.94	88.10	91.57	90.83	94.69	**99.16**
c2	80.79	55.8	80.17	92.71	84.27	81.35	**94.44**
c3	87.20	77.39	82.17	90.43	89.81	**97.75**	95.43
c4	78.05	**95.91**	87.43	89.76	82.82	95.83	94.48
c5	59.13	61.4	66.02	86.29	70.09	89.41	**93.95**
c6	90.21	90.81	91.13	90.57	91.65	95.60	**96.68**
c7	84.98	91.9	85.29	90.46	90.94	**94.44**	92.93
c8	89.40	77.14	85.23	92.48	91.11	94.92	**97.01**
c9	69.34	54.53	80.63	**89.96**	84.52	89.41	87.76
c10	53.14	75.48	79.67	87.27	82.09	94.19	**94.34**
OA(%)	86.24	84.29	85.95	90.26	89.20	94.58	**94.74**
AA(%)	77.63	76.63	76.54	90.15	80.44	92.76	**94.62**
Kappa(%)	79.08	76.86	78.65	89.72	83.65	91.84	**92.74**
time(s)	4328.16	4486.23	3002.40	**2049.36**	2897.49	6720.18	5187.03
Parameters	4.27×10^6	7.82×10^6	3.94×10^6	3.98×10^6	1.32×10^7	3.53×10^5	9.25×10^5

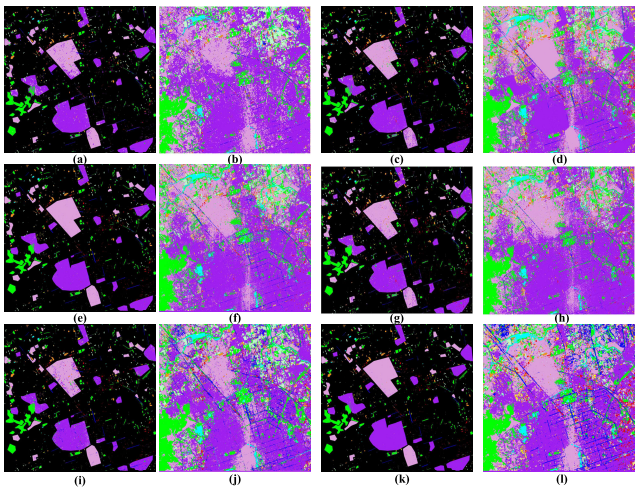


Fig. 7. Classification results of each algorithms in the Beijing dataset. Due to article space limitations, we only show six algorithms that have better results. Each algorithm has ground truth and overall results. (a) and (b) SCCA-Net [45]. (c) and (d) AHF-Net [2]. (e) and (f) MPCA-Net [46]. (g) and (h) TRLF-Net [47]. (i) and (j) GLT-Net [48]. (k) and (l) DSTD-Net.

For the Beijing area, Table VII and Fig. 7 show the quantitative experiment. The experimental results of different algorithms are analyzed as follows.

SCPF-ResNet [49] designs different modules to extract features from different input images. An adaptive spatial attention module (ASA-Module) and an adaptive channel attention module (ACA-Module) are designed to strengthen spatial features from targets with different sizes and spectral features among channels, which is beneficial for the feature extraction of remote sensing images. What's more, SCPF-ResNet adopts the idea of gradual fusion, which improves the performance of classification.

SCCA-Net [45] combines both sample enhancement and feature enhancement to improve classification accuracy. An adaptive neighborhood transfer sampling strategy is proposed to enhance samples. In terms of feature enhancement, they design a local SpA module and a global channel attention module, which is able to highlight the spatial information of the PAN image and the spectral advantage of the MS image. At the same time, it improves the difference between features through the interaction between the two modules. Therefore, compared with SCPF-Net, the SCCA-Net performs better in certain categories, such as c4: low vegetation (78.05% $\rightarrow$ 95.91%), c7: Dense high building (84.98% $\rightarrow$ 91.9%) and c10: white building (53.14% $\rightarrow$ 75.48%). However, the SCCA-Net lacks the strategy of gradual fusion, so SCCA-Net is inferior to SCPF-ResNet in terms of OA, AA, and Kappa.

AHF-Net [2] proposes an adaptive weighted IHS (AWIHS) strategy, which decreases the difference between MS and PAN images by sharing specific information. So AHF-Net performs better than SCPF-ResNet in most categories, such as c1: 84.07% $\rightarrow$ 88.10%, c4: 78.05% $\rightarrow$ 87.43% and c10: 53.14% $\rightarrow$ 79.67%, however, it has lower accuracy in the c8 category compared with SCPF-ResNet (89.40% $\rightarrow$ 85.23%). Since c8 accounts for a large proportion in the Beijing images, the SCPF-ResNet is superior to AHF-Net in terms of OA.

MPCA-Net [46] proposes an adaptive dilation rate selection strategy to adaptively select the dilation rate, solving the problem of category areas' excessive scale differences. Furthermore, it uses a new center pixel migration (CPM) strategy to increase the networks' stability. What's more, because of the different spatial and spectral features of PAN and MS, it designs different modules in different branches. Because of the CPM strategy, the accuracy of all categories increases a lot compared with SCPF-ResNet and the accuracy of categories that have low percentages is also very high, such as c1:91.57% and c10:87.27%. What's more, MPCA-Net has the fastest inference speed compared with all other algorithms, which is mainly owing to the simple structure of MPCA-Net.

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TRLF-Net [47] combines SpA and channel attention masks to enhance the PAN and MS images' respective advantages. What's more, it uses a deep dual-scale decomposition module to decompose the high-frequency and low-frequency components and minimize the essential difference. So TRLF-Net achieves a higher result in c6: (90.57% → 91.65%) and c7: (90.46% → 90.94%) compared with MPCA-Net. However, the TRLF-Net performs worse in c5: blue building (86.29% → 70.09%) and c2: Road (92.71% → 84.27%), which may result from the edge segmentation being fuzzy. The TRLF-Net isn't able to deal with the fuzzy edge information compared with MPCA-Net. The MPCA-Net's performance is better than the TRLF-Net.

GLT-Net [48] includes three parts: multiscale local spatial feature learning, global spectral feature learning, and global-local feature fusion classification. It is a network used for HSI and light detection and ranging (LiDAR) Data. It uses multiscale convolutional neural networks to learn local spatial features, while it uses a ViT to model global spectral dependencies. So GLT-Net has higher results in c3: (90.43% → 97.75%), c4 (89.76% → 95.83%) and c10 (87.27% → 94.19%) compared with MPCA-Net. GLT-Net combines the advantages of CNN and Transformer, so it behaves a better performance than MPCA-Net.

Our DSTD-Net takes the PAN and MS images' different advantages into account and proposes an SCA block to extract both MS and PAN images' common features and specific features. Considering the different advantages of MS and PAN, our SCA block uses SpA and spectral attention masks to guide the self-attention in the SCA block to extract the specific features. What's more, the attention mechanism in our SCA block extracts the global features that can distinguish the fuzzy pixels that make the classification results of different categories more balanced. So the OA, the AA, and the Kappa are all higher than other networks such as MPCA-Net, GLT-Net, and so on. We design a strengthened self-attention module enhanced by diff-attention, which can obtain the difference between two branches' specific features and improve it. Our DSTD-Net has higher overall precision than other methods that we mentioned before, and its accuracy in c1, c3, c5, c8, and c10 has improved considerably.

Our algorithm's running time is a little bit long since our attention mechanism needs more time than other algorithms. However, the parameters of our model are less than most algorithms in Table VII. In the architecture of the transformer, the point multiplication operation, used to generate attention maps, needs extra calculation while it doesn't have parameters. So, our network takes a long time to infer with fewer parameters.

As Table VIII shows, shanghai1 shows the results of training and testing on the Shanghai dataset, while shanghai2 shows the results of training on the Beijing dataset but testing on the Shanghai dataset. It can be seen that the outcome of shanghai2 is within an acceptable range. So, we think our model has a certain robustness.

H. Experimental Result With Huhhot Area Dataset, Xian Area Dataset, and Shanghai Area Dataset

Our DSTD-Net still has the highest AA, OA and Kappa in other datasets.

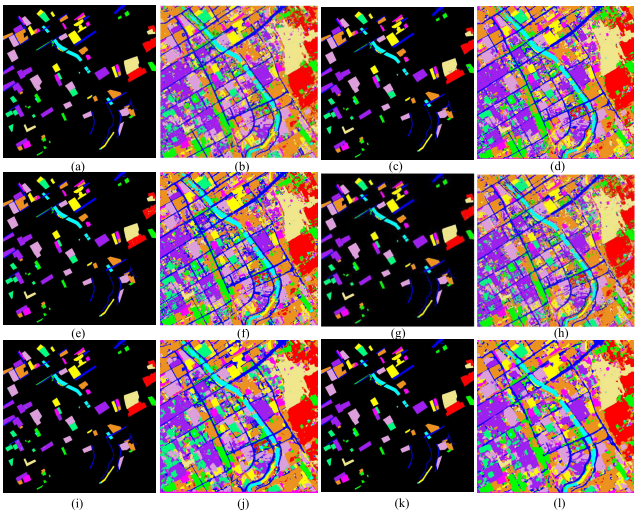


Fig. 8. Classification results of each algorithms in Huhhot dataset. Due to article space limitations, we only show six algorithms that have better results. Each algorithm has ground truth and overall results. (a) and (b) SCCA-Net [45]. (c) and (d) AHF-Net [2]. (e) and (f) MPCA-Net [46]. (g) and (h) TRLF-Net [47]. (i) and (j) GLT-Net [48]. (k) and (l) DSTD-Net.

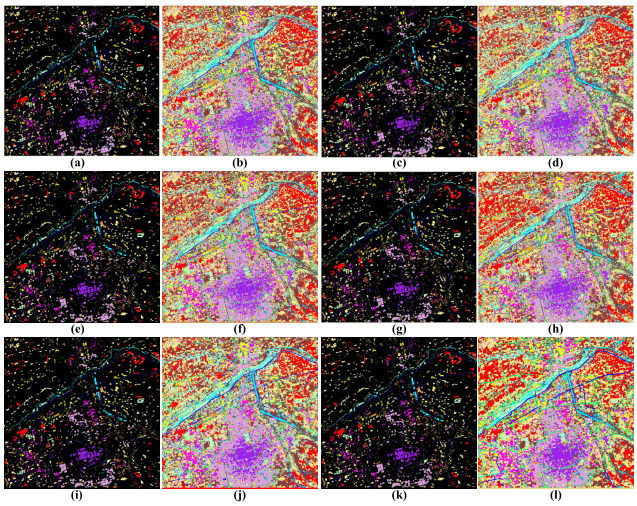


Fig. 9. Classification results of each algorithms in Xi'an dataset. Due to article space limitations, we only show six algorithms that have better results. Each algorithm has ground truth and overall results. (a) and (b) SCCA-Net [45]. (c) and (d) AHF-Net [2]. (e) and (f) MPCA-Net [46]. (g) and (h) TRLF-Net [47]. (i) and (j) GLT-Net [48]. (k) and (l) DSTD-Net.

For the Huhhot Area Dataset, the experimental result is shown in Table IX and Fig. 8. All in all, our DSTD-Net has higher accuracy in most categories compared with the other five methods, especially on c2 (Road), c7 (Farmland), and c9 (Building4). Categories c2, c7, and c9 have rich spectral and spatial information. Our DSTD-Net can extract the spectral and spatial features very well. The TRLF-Net has a better result in c5 (99.76% versus 99.52%) compared with the DSTD-Net, which because the farmland is intricate and the characteristics of different farmland in different areas are very different. The TRLF-Net has a representation fusion of the global and local features (RFGAL) measure to sort separate samples according to the neighborhood information.

For the results of the Xi'an area, Table X and Fig. 9 show quantitative and qualitative results.

We can see that c2 (Road1), c3 (Road2), c4 (Lowvegetation) and c7 (Tree) are very hardly distinguishable in all the categories by observing the confusion matrix. Overall,

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TABLE VIII
TRANSFER EXPERIMENT RESULTS BETWEEN THE BEIJING AREA DATASET AND THE SHANGHAI AREA DATASET (%)

	c1	c2	c3	c4	c5	c6	c7	c8	c9	c10	OA	AA	KAPPA
shanghai1	93.97	93.5	95.90	92.63	93.66	94.50	96.24	94.04	93.60	98.40	95.35	94.65	92.69
shanghai2	79.05	61.52	95.93	97.75	89.97	86.36	92.17	88.28	71.55	94.32	90.22	85.69	84.06

TABLE IX
QUANTITATIVE RESULT ON THE HUHHOT IMAGE

	SCPF -ResNet	SCCA -Net	AHF -Net	MPCA -Net	TRLF -Net	GLT -Net	DSTD -Net
c1	99.69	98.84	99.22	99.61	99.12	**99.83**	99.16
c2	97.21	94.72	94.35	96.95	98.10	96.67	**99.04**
c3	97.30	95.37	96.25	96.13	98.83	98.99	**99.13**
c4	98.48	96.24	96.81	97.63	99.26	97.49	**99.36**
c5	97.53	98.09	98.52	97.84	**99.76**	97.58	99.52
c6	97.64	96.12	97.23	96.20	99.40	99.09	**99.79**
c7	98.54	98.84	98.10	98.88	99.19	98.48	**99.85**
c8	98.65	96.57	97.83	97.47	99.39	**99.93**	99.31
c9	97.96	97.63	96.22	96.78	98.98	98.21	**99.73**
c10	97.61	96.93	96.15	96.79	98.96	**99.60**	99.24
c11	97.63	98.11	95.69	99.35	98.55	99.42	**99.60**
OA(%)	97.86	96.80	96.82	97.38	99.03	98.54	**99.42**
AA(%)	97.78	96.95	96.64	97.60	98.93	98.66	**99.43**
Kappa(%)	97.61	96.42	96.27	97.27	98.91	98.36	**99.34**
time(s)	222.23	386.28	412.18	542.28	**185.05**	241.78	292.37

TABLE X
QUANTITATIVE RESULT ON THE XI'AN IMAGE

	SCPF -ResNet	SCCA -Net	AHF -Net	MPCA -Net	TRLF -Net	GLT -Net	DSTD -Net
c1	99.57	98.75	99.02	99.39	**99.94**	99.93	99.78
c2	98.41	97.01	98.54	97.46	99.36	**99.63**	99.28
c3	97.08	96.33	93.12	96.06	**98.98**	97.62	98.55
c4	94.92	97.78	96.07	93.25	**98.99**	98.85	98.34
c5	93.93	89.56	94.49	92.59	98.03	99.46	**99.88**
c6	99.08	99.28	99.09	98.4	99.64	98.54	**99.83**
c7	98.91	96.30	95.93	97.95	99.26	97.54	**99.46**
c8	94.52	99.79	97.41	95.93	98.92	99.74	**99.93**
c9	96.85	96.55	97.89	98.97	98.98	99.70	**99.74**
c10	99.18	99.53	97.96	98.58	99.53	99.80	**99.89**
c11	99.04	96.79	98.14	99.02	99.53	98.47	**99.74**
c12	95.85	96.74	96.38	98.84	99.53	97.76	**99.58**
OA(%)	97.55	97.13	96.59	97.2	99.32	99.16	**99.64**
AA(%)	97.27	96.53	96.18	97.34	99.22	98.92	**99.49**
Kappa(%)	97.23	96.74	95.91	97.08	99.23	99.04	**99.59**
time(s)	507.55	2061.31	1252.16	1321.24	**412.18**	1598.42	1360.32

the DSTD-Net is superior to the other five methods for the evaluation indicators of AA, OA, and Kappa, which results from our SSDA block that can fetch the difference between specific features and make them purer. The Xi'an area has two kinds of roads: c2 and c3, which are so bumpy and complex for us to distinguish them. What's more, the proportion of c4 (bare land) is very low in the Xi'an area's labeled pixels. But in the DSTD-Net, our method can improve the feature's ability to characterize categories from both the data itself and feature extraction. However, the DSTD-Net doesn't perform in some difficulty categories, such as c1 (99.94% versus 99.78%), c3 (98.98% versus 98.55%), and c4 (98.99% versus 98.34%).

The results of the Shanghai area are shown in Table XI and Fig. 10.

In the Shanghai Image set, all the kinds of buildings are marked very well, but c9: red buildings account for a very small proportion in the images. So the c9 pixels belong to the difficult categories while c6, c7, c8, and c10 belong to simple categories. Our DSTD-Net has achieved higher results overall compared with other algorithms. However, our result is lower

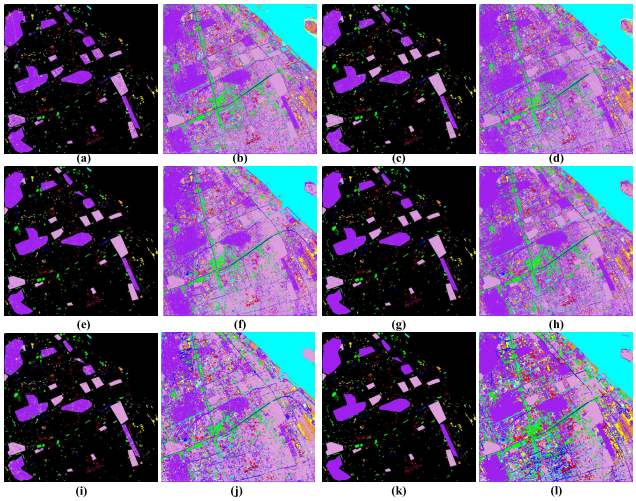


Fig. 10. Classification results of each algorithms in the Shanghai dataset. Due to article space limitations, we only show six algorithms that have better results. Each algorithm has ground truth and overall results. (a) and (b) SCCA-Net [45]. (c) and (d) AHF-Net [2]. (e) and (f) MPCA-Net [46]. (g) and (h) TRLF-Net [47]. (i) and (j) GLT-Net [48]. (k) and (l) DSTD-Net.

TABLE XI
QUANTITATIVE RESULT ON THE SHANGHAI IMAGE

	SCPF -ResNet	SCCA -Net	AHF -Net	MPCA -Net	TRLF -Net	GLT -Net	DSTD -Net
c1	86.37	83.49	83.71	92.37	89.26	91.37	**93.97**
c2	73.55	78.79	52.90	**93.68**	91.99	91.62	93.50
c3	95.68	89.55	77.90	91.28	88.83	85.03	**95.9**
c4	94.38	93.62	64.76	88.91	81.90	**97.05**	92.63
c5	88.09	71.74	80.28	89.27	88.93	**95.80**	93.66
c6	86.24	77.96	76.67	87.42	86.07	89.49	**94.50**
c7	91.11	85.44	76.66	90.10	92.50	88.00	**96.24**
c8	85.14	84.88	83.01	85.49	87.91	**94.49**	94.04
c9	77.58	92.35	65.61	94.57	93.21	**96.65**	93.60
c10	94.74	84.26	93.61	95.87	96.78	98.00	**98.40**
OA(%)	89.03	84.40	78.38	89.83	90.64	90.42	**95.35**
AA(%)	87.59	84.51	75.51	90.70	88.95	92.75	**94.65**
Kappa(%)	82.09	73.69	65.56	89.69	84.60	85.16	**92.69**
time(s)	2467.96	4937.96	3100.89	2317.10	**1187**	5283.08	3884.48

than MPCA-Net in c2 (93.68% versus 93.50%), and GLT-Net in c4 (97.05% versus 92.63%), c5 (95.80% versus 93.66%) and c9 (96.65% versus 93.60%). This is mainly because the MPCA-Net's application of CPM can better distinguish boundary pixels. GLT-Net can classify low_vegetation and building categories since it combines the advantages of CNN and Transformer, which is conducive to classifying features.

Despite the fact that our network doesn't perform best on some categories compared with other methods, the proposed DSTD-Net obtains the highest AA, OA, and Kappa in all the experiments compared with the other five algorithms.

V. CONCLUSION

We propose the DSTD-Net for PAN and MS image classification in this article. The experiments on the four datasets

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and the comparison with the other five algorithms are able to prove the effectiveness of our network. In our SSDA block, we design a diff-attention module that provides a new approach to remove the redundancy of specific features and help them be purer. What's more, our MGF strategy is helpful in strengthening advantageous features. However, our DSTD still has some problems remaining to be solved. For example, our network is not robust enough for edge details and spends extra time on model inference. In the near future, we will consider using the idea of multiscale to fetch edge details information and consider improving the speed of model inference with some improved self-attention modules.

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Lin Xu is currently pursuing the B.S. degree with the School of Artificial Intelligence, Xidian University, Xi'an, China.

His research interests include deep learning, computer vision, remote sensing image processing, and machine learning.



Hao Zhu (Member, IEEE) received the B.S. degree in physics and photoelectricity engineering and the Ph.D. degree in circuits and systems from Xidian University, Xi'an, China, in 2013 and 2019, respectively.

He is currently an Associate Professor with the School of Artificial Intelligence and the Key Laboratory of Intelligent Perception and Image Understanding of Ministry of Education, Xidian University. His research interests include deep learning, multimodal remote sensing image fusion and classification, and remote sensing object detection and tracking.



Licheng Jiao (Fellow, IEEE) received the B.S. degree from Shanghai Jiao Tong University, Shanghai, China, in 1982, and the M.S. and Ph.D. degrees from Xi'an Jiaotong University, Xi'an, China, in 1984 and 1990, respectively.

Since 1992, he has been a Professor with the School of Artificial Intelligence, Xidian University, Xi'an, where he is currently the Director of the Key Laboratory of Intelligent Perception and Image Understanding of the Ministry of Education of China. He is in charge of about 40 important scientific research projects and has authored more than 20 monographs and 100 papers in international journals and conferences. His research interests include image processing, natural computation, machine learning, and intelligent information processing.

Dr. Jiao is a fellow of the IEEE Xi'an Section Execution Committee, the Chairperson of the Awards and Recognition Committee, the Vice Board Chairperson of the Chinese Association of Artificial Intelligence, a Councilor of the Chinese Institute of Electronics, a Committee Member of the Chinese Committee of Neural Networks, and an Expert of the Academic Degrees Committee of the State Council.



Wenhao Zhao received the B.S. degree from Xidian University, Xi'an, China, in 2022, where he is currently pursuing the M.S. degree with the School of Artificial Intelligence.

His research interests include computer vision and remote sensing image processing.



Xiaotong Li received the B.S. degree in electronic information engineering from Harbin Engineering University, Harbin, China, in 2017. She is currently pursuing the Ph.D. degree with Xidian University, Xi'an, China.

Her main research interests include deep learning, computer vision, and remote sensing image analysis and understanding. She has participated in many international object tracking competitions.

Ms. Li received the First Prize in ECCV2018 VisDrone.



Biao Hou (Senior Member, IEEE) was born in Longxian, Shaanxi, China, in 1974. He received the B.S. and M.S. degrees in mathematics from Northwest University, Xi'an, China, in 1996 and 1999, respectively, and the Ph.D. degree in circuits and systems from Xidian University, Xi'an, in 2003.

Since 2003, he has been with the Key Laboratory of Intelligent Perception and Image Understanding of the Ministry of Education, Xidian University, where he is currently a Professor. His research interests include dictionary learning, deep learning, and synthetic aperture radar image processing.



Zhongle Ren (Member, IEEE) received the B.S. degree in electronic and information engineering and the Ph.D. degree in circuits and systems from Xidian University, Xi'an, China, in 2013 and 2020, respectively.

He is currently a Lecturer with the School of Artificial Intelligence, Xidian University. His research interests include machine learning, deep learning, and classification and object detection in remote sensing images.



Wenping Ma (Senior Member, IEEE) received the B.S. degree in computer science and technology and the Ph.D. degree in pattern recognition and intelligent systems from Xidian University, Xi'an, China, in 2003 and 2008, respectively.

Since 2006, she has been with the Key Laboratory of Intelligent Perception and Image Understanding of the Ministry of Education, Xidian University, where she is currently an Associate Professor. Her research interests include intelligent computing and image processing.

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